

Deformable Neural Radiance Fields for Object Motion Blur Removal

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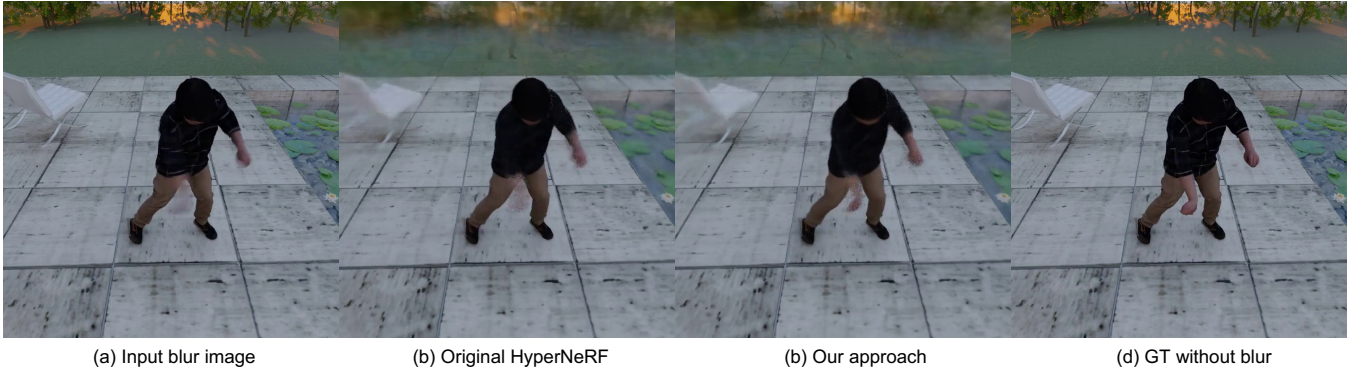


Figure 1: (a) Input image containing object motion blur, (b) HyperNeRF [Park et al. 2021] trained using the input image, (c) our approach controls blur removal by rendering with the latent hyperspace code from a neighboring frame with minimal motion blur, and (d) comparison with a ground truth image without object motion blur.

ABSTRACT

In this paper, we present a novel approach to remove object motion blur in 3D scene renderings using deformable neural radiance fields. Our technique adapts the hyperspace representation to accommodate shape changes induced by object motion blur. Experiments on Blender-generated datasets demonstrate the effectiveness of our method in producing higher-quality images with reduced object motion blur artifacts.

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1 INTRODUCTION

The advent of Neural Radiance Fields (NeRF) [Mildenhall et al. 2020] has led to a significant improvement in the quality of novel view synthesis for three-dimensional scenes. NeRF achieves this by continuously representing the color and density of 3D space as an implicit function and utilizing volume rendering to generate photorealistic images from arbitrary camera poses. However, obtaining high-quality NeRF representations requires careful image or video capture, as factors such as motion blur can significantly degrade the resulting synthesized views. Specifically, motion blur may occur when the relative position of the camera and the subject changes during the exposure time. Although existing research [Ma et al. 2021] [Lee et al. 2022] has successfully tackled the removal of camera motion blur, leading to sharper NeRFs, the challenge of eliminating object motion blur remains unresolved.

In this paper, we propose a deformable NeRF approach that improves HyperNeRF [Park et al. 2021] by focusing on the removal of object motion blur. Our technique adapts the hyperspace representation of HyperNeRF to accommodate shape changes induced by object motion blur, assigning distinct roles to the deformation field and hyperspace slices. This enables the control of global deformation and object motion blur, resulting in higher quality 3D scene renderings that are less affected by motion blur artifacts and extending the applicability of NeRF-based techniques to a wider

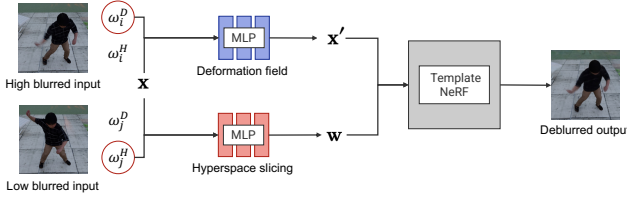


Figure 2: An overview of our proposed approach for object motion blur removal in HyperNeRF. Our method introduces a latent deformation code ω_i^D and a latent hyperspace code ω_i^H for each image. By substituting the latent hyperspace code of a neighboring frame with lower blur, our approach effectively mitigates object motion blur effects in the rendered images.

range of real-world scenarios. To evaluate the performance of our approach, we generated video datasets with and without object motion blur using Blender. We trained our model on the videos with object motion blur and evaluated the blur removal capability by comparing the deblurred output with the corresponding ground truth videos without object motion blur.

2 OUR APPROACH

In our approach, we address the challenges of object motion blur in HyperNeRF by assigning distinct roles to the deformation field, which manages global object transformations, and the hyperspace slices, which handle shape deformations caused by object motion blur. Our method extends the original HyperNeRF by introducing a latent hyperspace code ω_i^H for each image, complementing the existing latent deformation code ω_i^D and latent appearance code ψ_i . The core idea of our approach is to utilize the latent hyperspace code learned from frames with minimal object motion blur to render images with reduced blur artifacts. To accomplish this, we compute the Laplacian variance for each frame image and substitute the latent hyperspace code with that of a neighboring frame with lower variance values, effectively mitigating motion blur effects in the rendered images. Laplacian variance measures the image sharpness, with higher values indicating sharper images. By computing the Laplacian variance for each frame, we can identify images with minimal blur and select their corresponding latent hyperspace codes. This selection process helps render output images with reduced blur artifacts, resulting in more accurate representations.

3 EXPERIMENTS

We conducted both qualitative and quantitative evaluations to assess the performance of our proposed method. To evaluate the motion blur removal capability, we created a dataset of image pairs using Blender [Community 2018], with one image containing object motion blur and the other without. We simulated a setup of 100 fixed cameras surrounding the dynamic subject at 360 degrees, capturing images simultaneously. A total of 16 time frames were used. For our model, we employed 8-dimensional embeddings for the latent hyperspace code, the latent deformable code, and the latent appearance code. During training, we used images with object motion blur, while for testing, we compared the deblurred images

Table 1: Quantitative evaluation

Method	PSNR↑	SSIM↑	LPIPS↓
HyperNeRF	27.34	0.78	0.46
Ours	28.22	0.83	0.27

to their corresponding ground truth images without object motion blur to evaluate the effectiveness of our method in removing blur. For quantitative evaluation, we measured the visual quality of the resulting images using three widely accepted metrics: PSNR, SSIM and LPIPS metric. The quantitative results comparing our proposed approach with the original HyperNeRF are presented in Table 1.

4 CONCLUSION AND FUTURE WORK

In this paper, we presented a novel deformable NeRF approach for addressing object motion blur in HyperNeRF representations, resulting in higher quality 3D scene renderings with reduced blur artifacts. Through experiments using a Blender-generated dataset, our method demonstrated effectiveness in removing object motion blur and improving the visual quality of synthesized scenes. In future work, we plan to explore the use of flow-based NeRF [Du et al. 2021] techniques to simulate motion blur effects, which could potentially lead to further improvements in our approach for handling object motion blur in synthesized scenes.

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